Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Maida, James C M.S.		
Project Title:	Utilization of the SVS as an AR system for Mission Operations		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/15/2003	**End Date:**	11/30/2003
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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	Abstract Augmented reality is a technique whereby computer generated images are superimposed on live images for visual enhancement. Augmented reality can also be characterized as dynamic overlays when computer generated images are registered with moving objects in a live image. This technique has been successfully implemented, with low to medium levels of registration precision, in an NRA funded project entitled, Ã¢Â?Â?Improving Human Task Performance with Luminance Images and Dynamic OverlaysÃ¢Â?Â?. Future research is already being planned to also utilize a laboratory-based system where more extensive subject testing can be performed. However successful this might be, the problem will still be whether such a technology can be used with flight hardware. To answer this question, the Canadian Space Vision System (SVS) will be tested as an augmented reality system capable of improving human performance where the operation requires indirect viewing. This system has already been certified for flight and is currently flown on each shuttle mission for station assembly. Successful development and utilization of this system in a		

Task Description:	ground-based experiment will expand its utilization for on-orbit mission operations. Current research and development regarding the use of augmented reality technology is being simulated using ground-based equipment. This is an appropriate approach for development of symbology (graphics and annotation) optimal for human performance and for development of optimal image registration techniques. It is anticipated that this technology will become more pervasive as it matures. Because we know what and where almost everything is on ISS, this reduces the registration problem and improves the computer model of that reality, making augmented reality an attractive tool, provided we know how to use it. This is the basis for current research in this area. However, there is a missing element to this process. It is the link from this research to the current ISS video system and to flight hardware capable of utilizing this technology. This is the basis for this proposed Space Human Factors Engineering project, the determination of the display symbology within the performance limits of the Space Vision System that will objectively improve human performance. This utilization of existing flight hardware will greatly reduce the costs of implementation for flight. Besides being used onboard shuttle and space station and as a ground-based system for mission operational support, it also has great potential for science and medical training and diagnostics, remote learning, team learning, video/media conferencing, and educational outreach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Data on operator performance using augmented reality technology will be useful for ground based operations of remote systems using indirect vision (camera) systems such as surgery, construction, military, etc.
Task Progress:	This study demonstrated differences in alignment performance with respect to accuracy and speed for different augmented reality overlays and improved accuracy compared to alignment without overlays. The different overlays appeared to affect hand controller errors in different ways, suggesting that geometric differences among them may induce perceptual or cognitive difficulties in the operator. Additional research into these relationships may provide valuable insights into optimizing human interfaces for robotic controls.
Bibliography Type:	Description: (Last Updated: 02/27/2004)